

HONGLU JIANG, PH.D.

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Assistant Professor, Department of Computer Science and Software Engineering
Miami University

EDUCATION

- The George Washington University, Washington, DC May, 2021
Ph.D., Computer Science
- Qufu Normal University, Shandong, China June, 2012
M.S., Computer Applied Technology
- Qufu Normal University, Shandong, China July, 2009
B.E., Network Engineering

WORKING EXPERIENCE

- Assistant Professor (Sep. 2022-current)
Miami University, Oxford, Ohio
- Assistant Professor (Aug. 2021-Aug. 2022)
The University of Texas Rio Grande Valley, Brownsville, Texas
- Graduate Research Assistant (Jan. 2019-May. 2021)
The George Washington University, Washington, DC

RESEARCH INTERESTS

- Trustworthy AI
- Federated Learning
- Differential Privacy
- Wireless and Mobile Security
- IoT Networks and Edge Computing

GRANT AWARDS

- Multidisciplinary Pathways to Cybersecurity Professionals, NIST: PI, 10/01/2024-09/30/2026, \$199,850.
- Privacy-Preserving and Robust Federated Learning for IoT Systems, Miami University Research, Scholarship, and Artistry (RSA) Grant: PI, 06/01/2023-12/31/2024, \$12,942.
- Examining the Privacy Implications from the Use of Wearables in Human Performance Modeling for Occupational Purposes, Miami University Advanced Research Team (ART) grant: Co-PI, 02/01/2023-06/30/2024, \$59,339.

TEACHING

- Instructor of CYB 334 Network Security, Spring 2025 at Miami University.
- Instructor of CYB 235 Computer Network Design and Administration, Spring 2023, 2024 at Miami University.

- Instructor of CYB 234 System Administration and Scripting for Cybersecurity, Fall 2022, 2023, 2024 at Miami University.
- Instructor of CSCI 4341 IoT Principles & Hacking, Spring 2022 at UTRGV.
- Instructor of CSCI 3333 Algorithms & Data Structures, Spring 2022 at UTRGV.
- Instructor of CSCI 2344 Programming in Unix/Linux Environment, Fall 2021 at UTRGV.
- Teaching Assistant of CSCI 1311 Discrete Structures, Spring 2020 at GWU.
- Teaching Assistant of CSCI 4431 Computer Networks, Fall 2020 at GWU.
- Teaching Assistant of CSCI 4331 Cryptography, Fall 2019 at GWU.

STUDENTS SUPERVISION

Ph.D. Students (Co-advise):

- Nov. 2021-May 2024, Jianan Chen, Indiana University–Purdue University Indianapolis

Master Students:

- Sep. 2025-, Aymen Noor, Miami University
- Sep. 2025-May 2026, Dylan Middendorf, Miami University
- Mar. 2023-May 2026, Huy Tran, Miami University
- Sep. 2023-May 2025, Noah Greenbaum, Miami University
- Sep. 2023-May 2025, Ryan Holthouse, Miami University
- Sep. 2021- Aug. 2022, S. M. Sarwar, The University of Texas Rio Grande Valley

Undergraduate Students:

- Sep. 2024-, Liam McGuckin, Joseph Brocato, Jack McMaken, Sriram Ranganathan, Capstone project, Miami University
- Sep. 2024-, Caden Zeltne, Mason White, Michael Shyu, Sarah Staples, Capstone project, Miami University
- Sep. 2023-, Sriram Ranganathan, Miami University
- Sep. 2022-May 2023, Jenna Sayle, Miami University

PUBLICATIONS

Book

1. Jiguo Yu, Xiuzhen Cheng, Honglu Jiang, Dongxiao Yu, “Hierarchical Topology Control for Wireless Networks: Theory, Algorithms, and Simulation,” *CRC Press*, Feb 2018.

Journal Articles

2. Jianan Chen, Honglu Jiang, Qin Hu, “Utility-Enhanced Personalized Privacy Preservation in Hierarchical Federated Learning,” *IEEE Transactions on Mobile Computing*, 2025.
3. Jianchao Zheng, Honglu Jiang*, Liwen Yu, Samer Y. Khamaiseh, Mostafizur Rahman, “On the Performance of Uplink NOMA Short-packet Communication System,” *IEEE Transactions on Vehicular Technology*, 2024.

4. Honglu Jiang, Haotian Yu, Xiuzhen Cheng, Jian Pei, Robert Pless Jiguo Yu, "DP2-Pub: Differentially Private High-Dimensional Data Publication with Invariant Post Randomization," *IEEE Transactions on Knowledge and Data Engineering*, 2023.
5. Xiazhi Lai, Honglu Jiang*, Suman Bhunia and Huy Tran, "Reducing Latency in MEC Networks with Short-Packet Communications," *IEEE Transactions on Vehicular Technology*, 73(2), 3000-3004, 2023.
6. Jianan Chen, Qin Hu and Honglu Jiang, "Alliance Makes Difference? Maximizing Social Welfare in Cross-Silo Federated Learning," *IEEE Transactions on Vehicular Technology*, 73(2), 2786-2798, 2023.
7. Jianan Chen, Qin Hu, Honglu Jiang, "Strategic signaling for utility control in audit games," *Computers & Security*, 118, 102721, 2022.
8. Honglu Jiang*, S M Sarwar, Haotian Yu, Sheikh Arif Islam, "Differentially private data publication with multi-level data utility," *High Confidence Computing*, 2022.
9. Surendra Pathak, Sheikh Ariful Islam, Honglu Jiang, Lei Xu, Emmett Tomai, "A survey on security analysis of Amazon echo devices," *High Confidence Computing*, 2022.
10. Honglu Jiang, Jian Pei, Dongxiao Yu, Jiguo Yu, Bei Gong, Xiuzhen Cheng, "Applications of Differential Privacy in Social Network Analysis: A Survey," *IEEE Transactions on Knowledge and Data Engineering*, 35(1):108-27 2021.
11. Honglu Jiang, Jiguo Yu, Xiuzhen Cheng, Cheng Zhang, Bei Gong, Haotian Yu, "Structure-Attribute Based Social Network De-Anonymization with Spectral Graph Partitioning," *IEEE Transactions on Computational Social Systems*, 2021.
12. Biwei Yan, Jiguo Yu, Meihong Yang, Honglu Jiang, Zhiguo Wan, Lina Ni, "A Novel Distributed Social Internet of Things Service Recommendation Scheme Based on LSH Forest," *Personal and Ubiquitous Computing*, 1013-1026, 2021.
13. Cheng Zhang, Honglu Jiang, Xiuzhen Cheng, Feng Zhao, Zhipeng Cai, Zhi Tian, "Utility Analysis on Privacy-Preservation Algorithms for Online Social Networks: An Empirical Study," *Personal and Ubiquitous Computing*, 25: 1063-1079, 2021.
14. Lina Ni, Chao Li, Xiao Wang, Honglu Jiang, Jiguo Yu, "DP-MCDBSCAN: Differential Privacy Preserving Multi-core DBSCAN Clustering for Network User Data," *IEEE Access*, Vol. 6, 21053-21063, 2018.
15. Yuhong Sun, Ruinian Li, Honglu Jiang, Jianchao Zheng, Lina Ni, "Modeling and Analysis of Random Dense CSMA Networks," *EURASIP Journal on Wireless Communications and Networking*, 1-17 2018.
16. Kan Yu, Meng Gao, Honglu Jiang, Guangshun Li, "Multi-Leader Election in Dynamic Sensor Networks," *EURASIP Journal on Wireless Communications and Networking*, 1-14, 2017.
17. Feng Li, Jiguo Yu, Feng Zhao, Honglu Jiang, "A Novel Analysis of Delay and Power Consumption for Polling Schemes in the IoT," *Tsinghua Science and Technology*, Vol. 22, No. 4, 368-378, 2017.
18. Jiguo Yu, Honglu Jiang, Guanghui Wang, Qiang Guo, "Clustering-based energy-efficient broadcast tree in wireless networks," *International Journal of Computers Communications & Control*, Vol. 7 (4), 785-790, 2014.

Conference Papers

19. Pengyu Zhou, Daniel Chong, Honglu Jiang, Junjie Zhang and Rui Dai, "FedPDF-Guard: A Federated Learning Approach for Robust Detection of Malicious PDF Files," NAECON 2025 - IEEE National Aerospace and Electronics Conference, Dayton, OH, USA, 2025, pp. 1-6, doi: 10.1109/NAECON65708.2025.11235483.

20. Zhaowen Chen, Mingze Zhang, Caleb Mostyn, Honglu Jiang, Xianglong Feng, “Curse to Blessing: Leveraging Model Inversion Attacks to Detect Backdoor Attacks in Federated Learning,” *IEEE International Performance Computing and Communications Conference (IPCCC)*, 2024.
21. Lihua Song, Chenying Cai, Shuhua Wei, Rochishnu Banerjee, Xianglong Feng, and Honglu Jiang*, “Byzantine-Robust Federated Learning Based on Blockchain,” *The International Conference on Wireless Artificial Intelligent Computing Systems and Applications (WASA)*, 2024.
22. Lihua Song, Jing Li, Honglu Jiang*, Shuhua Wei, Yufei Guo, “Adaptive Edge-Level Personalization on Hierarchical Federated Learning,” *IEEE International Performance Computing and Communications Conference (IPCCC)*, 2023.
23. Jianan Chen, Qin Hu, Honglu Jiang, “Social Welfare Maximization in Cross-Silo Federated Learning,” *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022.
24. Zhiling Wang, Qin Hu, Minghui Xu, Honglu Jiang, “Blockchain-based Edge Resource Sharing for Metaverse,” *IEEE 19th International Conference on Mobile Ad Hoc and Smart Systems (MASS)*, 620-626, 2022.
25. Honglu Jiang, Yifeng Gao, S M Sarwar, Luis GarzaPerez, Mahmudul Robin, “Differential Privacy in Privacy-Preserving Big Data and Learning: Challenge and Opportunity,” *Silicon Valley Cybersecurity Conference*, 2021.
26. Cheng Zhang, Honglu Jiang*, Yawei Wang, Qin Hu, Jiguo Yu, Xiuzhen Cheng, “User Identity De-Anonymization Based on Attributes,” *International Conference on Wireless Algorithms, Systems, and Applications*, Springer, 458-469, 2019.
27. Cheng Zhang, Shang Wu, Honglu Jiang, Yawei Wang, Jiguo Yu, Xiuzhen Cheng, “Attribute-Enhanced De-Anonymization of Online Social Networks,” *International Conference on Computational Data and Social Networks*, Springer, 256-267, 2019.
28. Qi Sun, Jiguo Yu, Honglu Jiang, Yixian Chen, Xiuzhen Cheng, “De-anonymizing Scale-Free Social Networks by Using Spectrum Partitioning Method,” *Procedia Computer Science*, Vol. 147, 441-445, 2019.
29. Honglu Jiang, Jiguo Yu, Chunqiang Hu, Cheng Zhang, Xiuzhen Cheng, “SA Framework Based De-Anonymization of Social Networks,” *Procedia Computer Science*, Vol. 129, 358-363, 2018.
30. Jianchao Zheng, Jiguo Yu, Honglu Jiang, Yunchuan Sun, “Modeling and Analysis of Two-tier MIMO Heterogeneous Cellular Network”, *Procedia Computer Science*, Vol. 129, 364-367, 2018.
31. Lu Zhang, Jiguo Yu, Chunqing Hu, Honglu Jiang, Qi Sun, “Fine-grained access control with privacy support and network service optimization in ad hoc networks,” *Procedia Computer Science*, Vol. 129, 372-374, 2018.
32. Yuhong Sun, Tianyi Song, Honglu Jiang, Jianchao Zheng, “Modeling of Random Dense CSMA Networks,” *International Conference on Wireless Algorithms, Systems, and Applications*, 71-82, 2017.

PROFESSIONAL ACTIVITIES

Selected Editorial Board Member

- Editor Board, Intelligent Information Management (IIM)

Guest Editor

- High Confidence Computing

- Special Issue on “Advances in 5G/6G-enabled Crowd Sensing IoT for *Digital Communications and Networks*, 2022
- Special Issue on “Advances in Sensing, Imaging, and Computing for Autonomous Driving” for *Sensors*, 2022

Paper Reviewer

- IEEE Internet of Things Journal
- IEEE Transactions on Knowledge and Data Engineering
- IEEE Transactions on Wireless Communications
- IEEE Transactions on Industrial Informatics
- IEEE Transactions on Vehicular Technology
- IEEE Transactions on Neural Networks and Learning Systems
- Computers & Security
- IEEE Transactions on Computational Social Systems
- IEEE Transactions on Network Science and Engineering
- ACM Transactions on Cyber-Physical Systems
- EURASIP Journal on Wireless Communications and Networking
- Frontiers of Information Technology & Electronic Engineering
- Digital Communications and Networks
- Computer Communications
- High Confidence Computing
- International Journal of Intelligent Systems
- Complexity
- Wireless Communications and Mobile Computing
- CCF Transactions on Networking
- Informs Journal on Computing
- IEEE INFOCOM 2022, 2023, 2024

Program Committee Member

- IEEE Vehicular Technology Conference (VTC) 2023
- ICCCN 2023, 2024
- ACM SIGKDD 2022
- REU Research in Networking and Systems (REUNS) 2022
- IIKI 2017

Panelist

- NSF Panelist, 2022, 2023, 2024

Membership

- Senior Member, IEEE

PRESENTATIONS AND INVITED TALKS

- “From Data to Models: Advancing Privacy through Differential Privacy,” Data Science team at Cincinnati Children’s Hospital, Nov. 2025. Invited Talk
- “Adaptive Edge-Level Personalization on Hierarchical Federated Learning,” IEEE International Performance Computing and Communications Conference (IPCCC), Anaheim, California, 2023. Conference Presentation
- “On Using Differential Privacy,” Department of Computer Science, Saint Joseph’s University, December 2022. Invited Talk
- “Women in CyberSecurity,” CyberSecurity Club, Miami University, Nov 2022. Invited Talk, Panelist
- “Blockchain-based Edge Resource Sharing for Metaverse,” IEEE MASS, Denver, USA, 2022. Conference Presentation